Unit 13

Linear Transformations

Unit Goals

- Be able to explain what is meant by a linear transformation.
- Be able to explain why the derivative and integral operators are linear transformations..
- Be able to find the matrix representation of a linear transformation.
- Be able to relate the eigenvalue and singular value decompositions to the search for a good basis.
- To be able to apply the concepts of bases and orthogonality to functions.

Unit 13.1

Matrices as mappings.

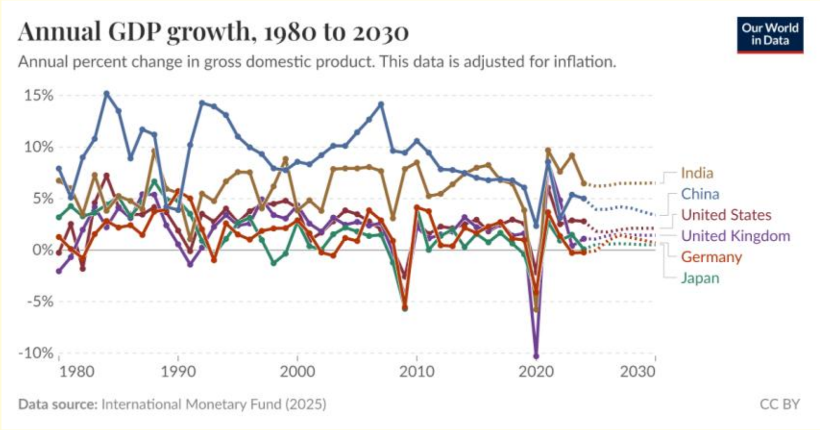
What do matrices "mean"?

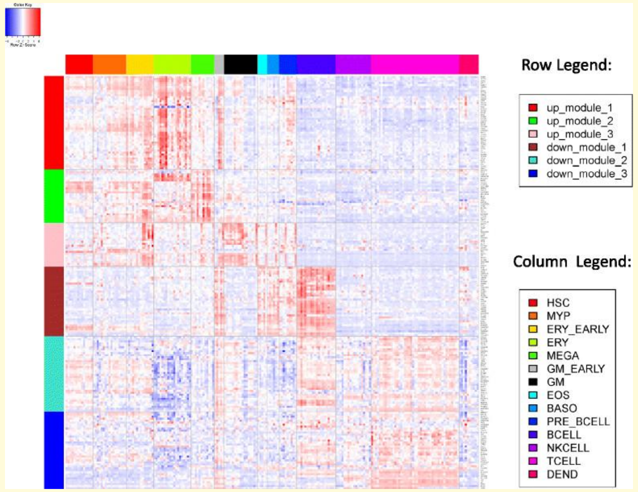
- A matrix may represent the coefficients of the linear system—a representation of the system of equations for a model.

$$A\vec{x} = \vec{b} \rightarrow Given\ A\ and\ \vec{b}, solve\ for\ \vec{x}.$$

$$A\vec{x}_i = \vec{x}_{i+1} \rightarrow Given\ A\ and\ \vec{x}_0, solve\ for\ \vec{x}_k$$

- A matrix may represent data:





What do matrices "mean"?

- In a dynamic system, the matrix creates a new vector from an input:

$$Given\ \vec{x}_i, A\vec{x}_i = \vec{x}_{i+1}$$

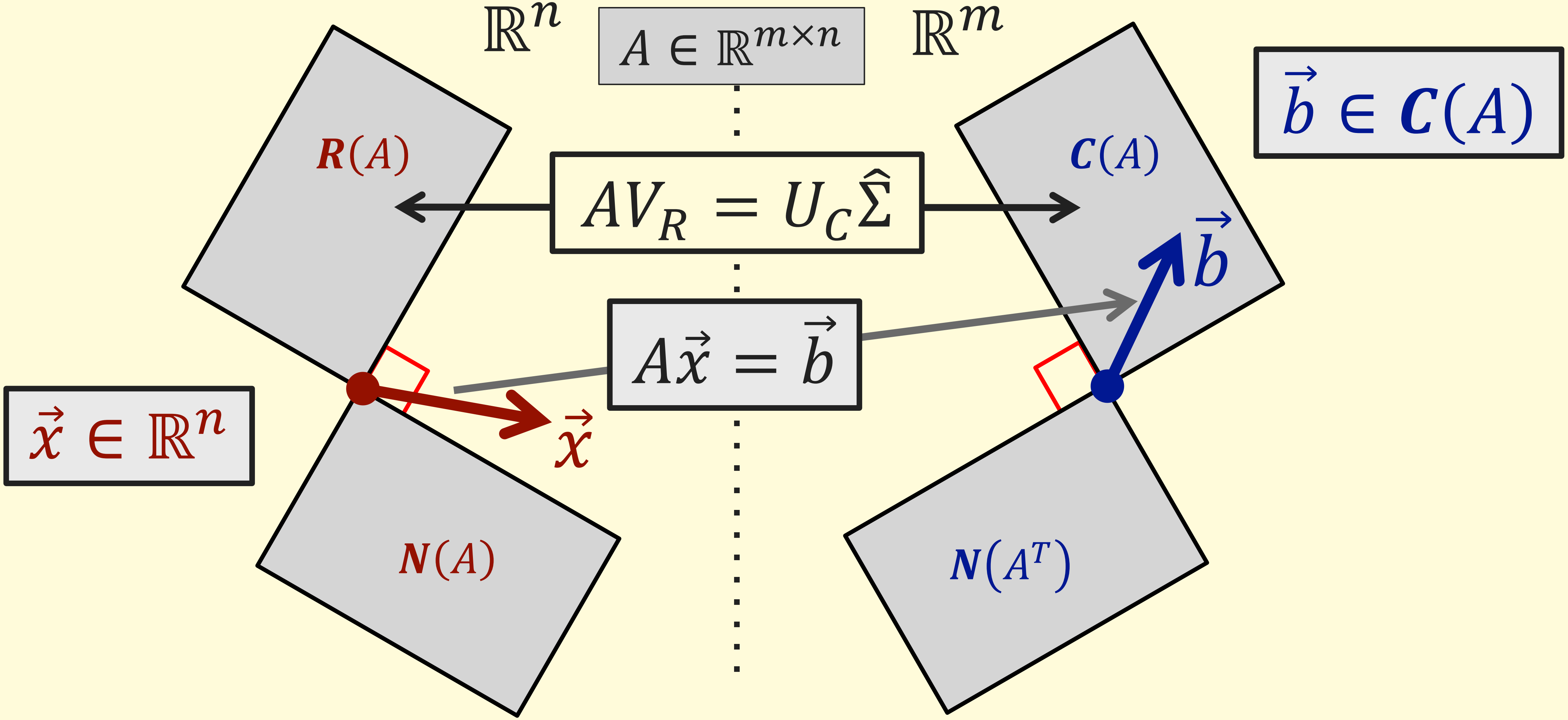
- In a system of equations, the RHS is "created" from the variables:

$$Given\ \vec{x}, A\vec{x} = \vec{b}$$

- **A matrix can represent a way to transform an input vector into some output vector, a "mapping":**

$$Given\ \vec{x}, A\vec{x} = \vec{y} \leftrightarrow A\colon\ \vec{x} \to \vec{y}$$

The big picture: A as a "mapping".

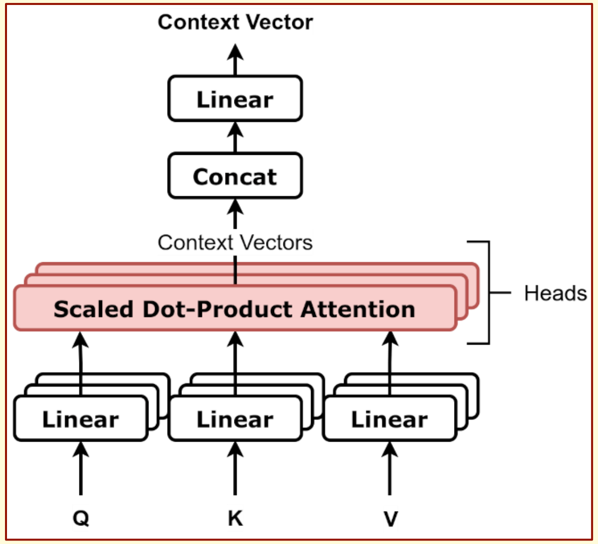


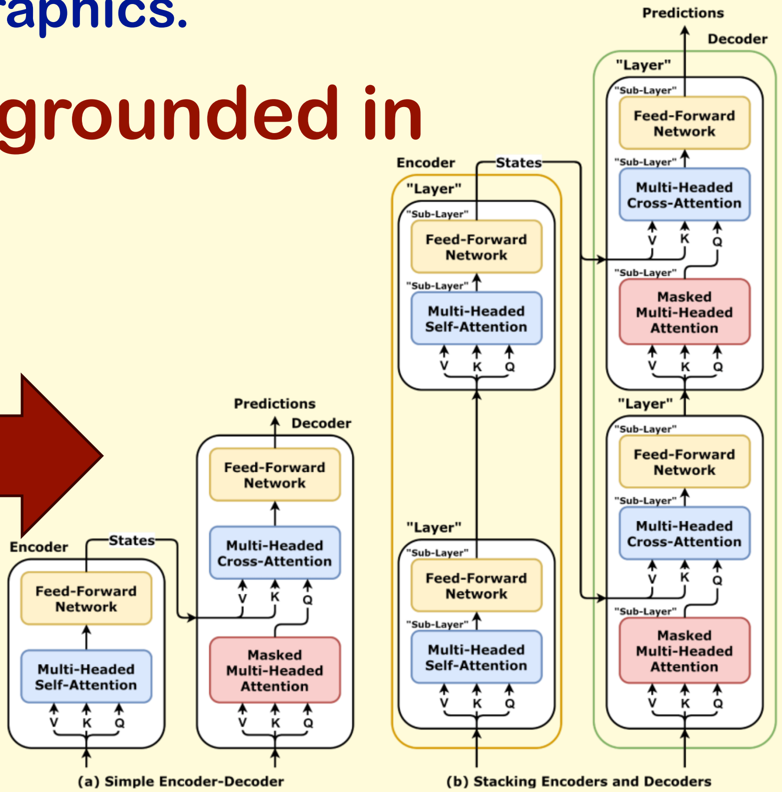
Why does this matter?

- In many applications we may wish to transform data:
  - Signal processing, image manipulation, computer graphics.
- **Modern Machine Learning methods are grounded in transformations!**

$$\boxed{\begin{aligned} W_Q \vec{x}_q &= \vec{q} \\ W_K \vec{x}_k &= \vec{k} \\ W_V \vec{x}_v &= \vec{v} \end{aligned}}$$

$$\boxed{A(Q,K,V) = f_{SM}\left(\frac{Q^T K}{\sqrt{\dim(K)}}\right) V}$$





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Unit 13.2

Generalized Vector Spaces

Abstract vector spaces.

- Vectors have two well-defined operations: addition and scalar multiplication.
  - In vector addition, we add corresponding terms. Vector addition is associative, commutative and has both an identity and an inverse:
$$\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}, \qquad \vec{u} + \vec{v} = \vec{v} + \vec{u},$$
$$\exists \vec{0}: \vec{u} + \vec{0} = \vec{u}, \forall \vec{u}, \qquad \forall \vec{u}, \exists - \vec{u}: \vec{u} + (-\vec{u}) = \vec{0}$$
  - In scalar multiplication, we multiply all elements of the vector by the same value. Scalar multiplication is distributive with vector addition and is compatible with “regular” scalar multiplication:
$$k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}, \qquad (k + l)\vec{u} = k\vec{u} + l\vec{u},$$
$$(kl)\vec{u} = k(l\vec{u}), \qquad 1\vec{u} = \vec{u}, \forall \vec{u}$$
- Vector spaces are closed with respect to these operations:
$$\vec{u} + \vec{v} \in V, \forall \vec{u}, \vec{v} \in V, \qquad k\vec{v} \in V, \forall \vec{v} \in V, \forall k \in \mathbb{R}$$
- Other mathematical objects can be worked with in the same way!

Abstract vector spaces.

- **If:** we can define a set of mathematical objects, V, that:
  - has two well-defined operations, addition and scalar multiplication,
  - where vector addition is associative, commutative and has both an identity and an inverse:
$$\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}, \qquad \boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u},$$
$$\exists \boldsymbol{0} \in V: \boldsymbol{u} + \boldsymbol{0} = \boldsymbol{u}, \forall \boldsymbol{u} \in V, \qquad \forall \boldsymbol{u} \in V, \exists -\boldsymbol{u} \in V: \boldsymbol{u} + (-\boldsymbol{u}) = \boldsymbol{0}$$
  - scalar multiplication is distributive with vector addition and is compatible with "regular" scalar multiplication:
$$k(\boldsymbol{u} + \boldsymbol{v}) = k\boldsymbol{u} + k\boldsymbol{v}, \qquad (k + l)\boldsymbol{u} = k\boldsymbol{u} + l\boldsymbol{u}$$
$$(kl)\boldsymbol{u} = k(l\boldsymbol{u}), \qquad 1\boldsymbol{u} = \boldsymbol{u}, \forall \boldsymbol{u} \in V$$
  - and that is closed with respect to these operations:
$$\boldsymbol{u} + \boldsymbol{v} \in V, \forall \boldsymbol{u}, \boldsymbol{v} \in V, \qquad k\boldsymbol{u} \in V, \forall \boldsymbol{u} \in V, \forall k \in \mathbb{R}$$
- **Then:** we can call V a vector space, and the elements of V are vectors.

Matrix space.

- Consider the set of all $m \times n$ matrices $\mathbb{R}^{m\times n}$.
- We can add matrices to get another matrix of the same size

$$A + B \in \mathbb{R}^{m\times n}, \forall A, B \in \mathbb{R}^{m\times n}$$

$$A + (B + C) = (A + B) + \mathrm{C}, \qquad A + B = B + A,$$

- We can define a zero matrix and an additive inverse:

$$\exists 0 \in \mathbb{R}^{m\times n}: A + 0 = A, \forall A \in \mathbb{R}^{m\times n}, \qquad \forall A \in \mathbb{R}^{m\times n}, \exists -A \in \mathbb{R}^{m\times n}: A + (-A) = 0$$

- We can multiply matrices by a scalar to get another matrix of the same size

$$kA \in \mathbb{R}^{m\times n}, \forall A \in \mathbb{R}^{m\times n}, \forall k \in \mathbb{R}$$

$$k(A + B) = kA + kB, \qquad (k + l)A = kA + lA$$

$$(kl)A = k(lA), \qquad 1A = A, \forall A \in \mathbb{R}^{m\times n}$$

- The set of all matrices of a given size is a vector space!

Function space.

- Consider the set of all functions of a single variable, $F = \{f(x)\}$.
- We can add functions to get another function of a single variable:
$$f(x) + g(x) = h(x) \to h(x) = (f + g)(x)$$
- We can multiply functions by a scalar to get another function of a single variable:
$$kf(x) = g(x) \to g(x) = (kf)(x)$$
- Function addition: group like terms and add coefficients:
  - Obeys all the rules; the zero function is $f(x) = 0$.
- Scalar multiplication: multiply all terms by the same scalar:
  - Obeys all the rules.
- **The set of all functions of a single variable is a vector space!**

Polynomial space.

- Consider the set of all functions of a single variable that can be written as a polynomial of degree n or less.:

$$P^n = \{f(x) \mid f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n\, x^n\}$$

- When we add polynomials, we get another polynomial of no greater degree:

$$f(x) + g(x) \in P^n, \forall f(x), g(x) \in P^n$$

- When we scale polynomials, we get another polynomial of the same degree:

$$kf(x) \in P^n, \forall f(x) \in P^n, \forall k \in \mathbb{R}$$

  → **The set of polynomials is closed with respect to addition and scalar multiplication.**

- The set of all polynomial functions of a single variable is a vector space; it is a subspace of function space!

Basis and Dimension with Generalized Vector Spaces

- Recall that a basis is a linearly independent spanning set for a vector space.
- A set $\{\vec{v}_1, \quad \vec{v}_2, \quad \cdots \quad \vec{v}_n\}$ is a basis for a vector space V if (and only if):
  - The set is linear independent; AND
  - Every vector in V can be written as a linear combination of the vectors $\vec{v}_i$, or:

$$V = span(\{\vec{v}_1, \quad \vec{v}_2, \quad \cdots \quad \vec{v}_n\})$$

- **The dimension of a vector space is the number of elements in any basis for the space.**
- What are bases for some general vector spaces?

Basis and dimension of $\mathbb{R}^{m\times n}$

- Each basis "vector" must be an m-by-n matrix; one basis is the set of matrices E_{ij} with a single "1" at position ij and zeros everywhere else.
- There are $m \times n$ matrices of this type, so $\dim(\mathbb{R}^{m\times n}) = mn$.

- Using $\mathbb{R}^{2\times 3}$ as an example:

$$\left\{\begin{bmatrix}1 & 0 & 0\\0 & 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 1 & 0\\0 & 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0 & 1\\0 & 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0 & 0\\1 & 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0 & 0\\0 & 1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0 & 0\\0 & 0 & 1\end{bmatrix}\right\}$$

  - The set is obviously linear independent.
  - The set is a spanning set:

$$\begin{bmatrix}a_{11} & a_{12} & a_{13}\\a_{21} & a_{22} & a_{23}\end{bmatrix} = a_{11}\begin{bmatrix}1 & 0 & 0\\0 & 0 & 0\end{bmatrix} + a_{12}\begin{bmatrix}0 & 1 & 0\\0 & 0 & 0\end{bmatrix} + a_{13}\begin{bmatrix}0 & 0 & 1\\0 & 0 & 0\end{bmatrix} + a_{21}\begin{bmatrix}0 & 0 & 0\\1 & 0 & 0\end{bmatrix} + a_{22}\begin{bmatrix}0 & 0 & 0\\0 & 1 & 0\end{bmatrix} + a_{23}\begin{bmatrix}0 & 0 & 0\\0 & 0 & 1\end{bmatrix}$$

  - The dimension of $\mathbb{R}^{2\times 3}$ is 6.

Basis and dimension of P^n

- Each basis "vector" must be a polynomial of degree n or less; one basis is the set of functions $\left\{x^i, \quad i = [0, n]\right\}$.
- There are $n + 1$ functions of this type, so $\dim(P^n) = n + 1$.

- Using P^3 as an example:

$$\{1, \quad x, \quad x^2, \quad x^3\}$$

  - The set is obviously linear independent.
  - The set is a spanning set:

$$p(x) = a_0(1) + a_1 x + a_2 x^2 + a_3 x^3$$

  - The dimension of P^3 is 4.

Basis and dimension of function space?

- Any continuous function on a closed domain can be uniformly approximated as closely as desired by a polynomial (Stone-Weierstrass Theorem).
- **But there is no bound on the degree!**
- What is the dimension of the vector space of polynomials of ***any*** degree?
- Basis:

$$\{1, \quad x, \quad x^2, \quad x^3, \quad \cdots\}$$

- This set is countable but infinite (just like the natural numbers).

$$\dim(P) = \infty$$

  - Formally, $\dim(P) = \aleph_0$, the cardinality of the natural numbers
- In many applications, we approximate function space by using a finite basis (for example, polynomials up to degree n).

Coordinate vectors.

- Any element of a vector space can be written as a linear combination of basis vectors.

$$V = span(\{\vec{v}_1, \quad \vec{v}_2, \quad \cdots \quad \vec{v}_n\}) \rightarrow \vec{x} = \sum_{i=1}^{n} a_i \vec{v}_i \,, \forall \vec{x} \in V$$

- Define a coordinate vector to be the ordered list of coefficients:

$$[\vec{x}] = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

- In this way, we can write any generalized vector in the same form as a “traditional” vector!

Coordinate vectors.

- **Example 1, a vector and basis from $\mathbb{R}^{2\times 2}$:**

$$A = \begin{bmatrix} 2 & 3 \\ 6 & 4 \end{bmatrix}, \qquad S = \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}$$

$$\begin{bmatrix} 2 & 3 \\ 6 & 4 \end{bmatrix} = 4\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + 2\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} - 3\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} - 1\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow [A]_S = \begin{bmatrix} 4 \\ 2 \\ -3 \\ -1 \end{bmatrix}$$

- **Example 2, a vector and basis from P^2:**

$$f(x) = 5 - x + 3x^2, \qquad L = \left\{1, \quad x, \quad \frac{1}{2}(3x^2 - 1)\right\}$$

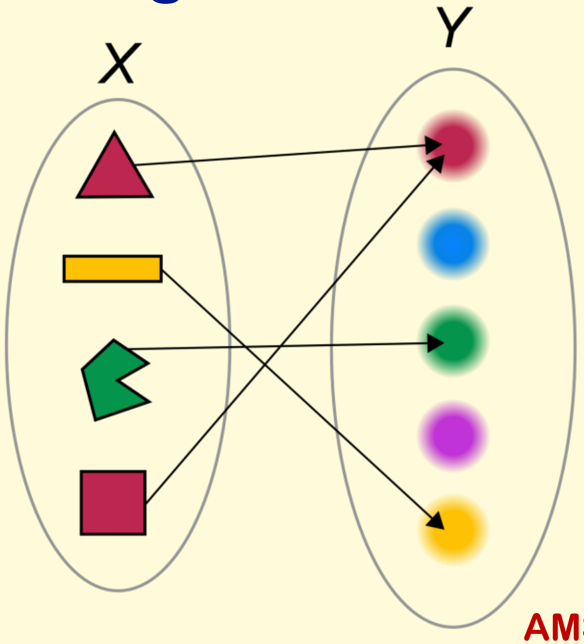
$$5 - x + 3x^2 = 6(1) - 1(x) + 2\left(\frac{1}{2}(3x^2 - 1)\right) \rightarrow [f(x)]_L = \begin{bmatrix} 6 \\ -1 \\ 2 \end{bmatrix}$$

Unit 13.3

Linear Mappings

What is a mapping?

- A mapping (map) is simply a rule that uniquely associates any object from some input set with another object from some output set; it can also be called a function.
- E.g.: a rule to map a uniformly colored shape onto the set of colors:



Important aspects:

- The input set is called the **domain**.
- The rule must apply to every element in the domain.
- There can not be any ambiguity in the output.

$$\forall \vec{x} \in X, \exists \vec{y} \in Y: \vec{y} = F(\vec{x})$$
$$F(\vec{x}) = \vec{y}_1, F(\vec{x}) = \vec{y}_2 \rightarrow \vec{y}_1 = \vec{y}_2$$

- The parent set for the output is called the codomain.
- The actual set of outputs is called the image or range.

Understanding mappings with a linear system.

For the system: $A\vec{x} = \vec{b}$

- What space does $\vec{x}$ come from?
- What space does $\vec{b}$ come from?
- Can one value of $\vec{b}$ give different values of $\vec{x}$ as a solution?
- Can different values of $\vec{b}$ give the same value of $\vec{x}$ as a solution?
- Do all possible values of $\vec{b}$ have a well-defined value of $\vec{x}$ as a solution?
- Is every possible value of $\vec{x}$ the solution for some value of $\vec{b}$?

- Do your answers depend on the properties of A (such as size and rank)?

A linear system as a mapping!

- We can think of A as *acting on* $\vec{x}$ to give $\vec{b}$.
- The matrix A *maps* the vector $\vec{x}$ onto the vector $\vec{b}$.
- For an $m \times n$ matrix, $\vec{x}$ must come from $\mathbb{R}^n$ and $\vec{b}$ must come from the column space of A, a subspace of $\mathbb{R}^m$.

- What space does $\vec{x}$ come from?
- What space does $\vec{b}$ come from?
- Do your answers depend on the properties of A?

A linear system as a mapping!

- **The matrix A maps the vector $\vec{x} \in \mathbb{R}^n$ onto the vector $\vec{b} \in C(A) \subseteq \mathbb{R}^m$.**
- **Every has $\vec{x}$ to give an answer $\vec{b}$ and only one:**

$$\forall \vec{x} \in X, \exists \vec{y} \in Y: \vec{y} = F(\vec{x})$$
$$F(\vec{x}) = \vec{y}_1, F(\vec{x}) = \vec{y}_2 \rightarrow \vec{y}_1 = \vec{y}_2$$

- **Can two different values of $\vec{b}$ give the same value of $\vec{x}$ as a solution?**
- **Is every possible value of $\vec{x}$ the solution for some value of $\vec{b}$?**
- **Do your answers depend on the properties of A?**

A linear system as a mapping!

- The matrix A maps *any* vector $\vec{x} \in \mathbb{R}^n$ onto one vector $\vec{b} \in C(A) \subseteq \mathbb{R}^m$.
- Some systems have multiple solutions → many $\vec{x}$ give the same $\vec{b}$.
- Some systems have no solution → there is no $\vec{x}$ giving the desired $\vec{b}$.

- Can one value of $\vec{b}$ give different values of $\vec{x}$ as a solution?
- Do all possible values of $\vec{b}$ have a well-defined value of $\vec{x}$ as a solution?
- Do your answers depend on the properties of A?

A linear system as a mapping!

- All mappings → every $\vec{x}$ from $\mathbb{R}^n$ gives some $\vec{b}$ from $\mathbb{R}^m$.
  - $A\vec{x} = \vec{b}$: Matrix multiplication always works!
- **One-to-one** mappings → each $\vec{x}$ gives a different $\vec{b}$.
  - $A\vec{x} = \vec{b}$: Null space is just the origin; row space is all of $\mathbb{R}^n$.
- **Onto** mappings → every $\vec{b}$ can be constructed from some $\vec{x}$.
  - $A\vec{x} = \vec{b}$: Left null space is just the origin; column space is all of $\mathbb{R}^m$.
  - Image (or range) is the entire codomain.
- **One-to one correspondence** → both “one-to-one” and “onto”.
  - $A\vec{x} = \vec{b}$: Full rank (square and non-singular); matrix has an inverse.
  - There is a well-defined inverse mapping: $\vec{b} = A^{-1}\vec{x}$.

Surjective

Injective

Bijective

Linearity: What is a linear map?

- A linear map or linear transformation is a rule that uniquely associates one vector (of any type) to another and ***which preserves linear combinations.***
- Consider a rule to map a vector from vector space U to a vector from vector space V, $F\colon V \to U$.
- To be a linear map, the rule must satisfy:

$$\forall \vec{v} \in V, \exists!\, \vec{u} \in U\colon \vec{u} = F(\vec{v})$$

$$F(k\vec{v}_1 + l\vec{v}_2) = kF(\vec{v}_1) + lF(\vec{v}_2), \qquad \forall \vec{v}_1, \vec{v}_2 \in V$$

Kernel and Range

- The ***kernel*** of a linear map is the set of input values that map to zero:

$$kernel(F: V \to U) = \left\{\vec{v} \in V : F(\vec{v}) = \vec{0}\right\}$$

  - Comparable to the null space of a matrix.
  - For a one-to-one mapping, the kernel is only the zero vector.

- The ***range*** of a linear map is the set of possible output values:

$$range(F: V \to U) = \{\vec{u} \in U : \exists \vec{v} \in V, \vec{u} = F(\vec{v})\}$$

  - Comparable to the column space of a matrix.
  - For an onto mapping, the range is the entire codomain.

Matrix multiplication is a linear map

- Consider an m-by-n matrix A:

$$\forall \vec{v} \in \mathbb{R}^n, \exists! \, \vec{u} \in \mathbb{R}^m : \vec{u} = A\vec{v}$$

$$A(k\vec{v}_1 + l\vec{v}_2) = kA\vec{v}_1 + lA\vec{v}_2, \qquad \forall \vec{v}_1, \vec{v}_2 \in \mathbb{R}^n$$

$F(\vec{v}) = A\vec{v}$ is a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$.

- In this case, the kernel of F is identical to the null space of A, and the range of F is identical to the column space of A.

Derivatives are linear maps!

- Consider the set of all smooth functions of a single variable, C^∞:

$$\forall f(x) \in C^\infty, \exists!\, f'(x) \in C^\infty : f'(x) = \frac{d}{dx} f(x)$$

$$\frac{d}{dx}\big(kf(x) + lg(x)\big) = k\frac{d}{dx} f(x) + l\frac{d}{dx} g(x) = kf'(x) + lg'(x),$$

$$\forall f(x), g(x) \in C^\infty$$

$f'(x) = \frac{d}{dx} f(x)$ is a linear map from C^∞ to C^∞.

- Question: If the derivative map acts on P^n, what is the codomain?

Indefinite integrals are linear maps!

- Consider the set of all smooth functions of a single variable, C^∞:

$$\forall f(x) \in C^\infty, \exists!\, F(x) \in C^\infty : F(x) = \int f(x)dx\,, F(0) = 0$$

$$\int \big(kf(x) + lg(x)\big) = k\int f(x)dx + l\int g(x)dx = kF(x) + lG(x),$$

$$\forall f(x), g(x) \in C^\infty$$

$F(x) = \int f(x)dx\,, F(0) = 0$ is a linear map from C^∞ to C^∞.

- Note, $F(0) = 0$ implies the constant of integration is zero. This is required to ensure uniqueness and that $f(x) = 0$ maps to $F(x) = 0$.
- Question: If the integral map acts on P^n, what is the codomain?

Linear maps can be represented by matrices

- Coordinate vectors link generalized vector spaces with ordered lists:

$$V = span(\{\vec{v}_1, \quad \vec{v}_2, \quad \cdots \quad \vec{v}_n\}) \rightarrow \vec{x} = \sum_{i=1}^{n} a_i \vec{v}_i \,, \forall \vec{x} \in V \rightarrow [\vec{x}] = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

- Consider a linear map, $F: V \rightarrow U$:

$$F(\vec{x}) = F\left(\sum_{i=1}^{n} a_i \vec{v}_i\right) = \sum_{i=1}^{n} a_i F(\vec{v}_i) \,, \forall \vec{x} \in V$$

- The column view of a linear system gives:

$$\sum_{i=1}^{n} a_i F(\vec{v}_i) = [F(\vec{v}_1) \quad F(\vec{v}_2) \quad \cdots \quad F(\vec{v}_n)] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = [F][\vec{x}]$$

- The matrix representation of a linear map is a matrix whose columns are the mapped basis!

Linear maps can be represented by matrices

- **Coordinate vectors link generalized vector spaces with ordered lists:**

$$V = span(\{\vec{v}_1, \quad \vec{v}_2, \quad \cdots \quad \vec{v}_n\}) \rightarrow \vec{x} = \sum_{i=1}^{n} a_i \vec{v}_i \,, \forall \vec{x} \in V \rightarrow [\vec{x}] = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

- **Consider a linear map, $F\colon V \rightarrow U$:**

$$F(\vec{x}) = F\left(\sum_{i=1}^{n} a_i \vec{v}_i\right) = \sum_{i=1}^{n} a_i F(\vec{v}_i) \,, \forall \vec{x} \in V$$

- **$F(\vec{x})$ can also be represented as a coordinate vector:**

$$U = span(\{\vec{u}_1, \quad \vec{u}_2, \quad \cdots \quad \vec{u}_n\}) \rightarrow F(\vec{x}) = \sum_{i=1}^{m} b_i \vec{u}_i \,, \forall \vec{x} \in V \rightarrow [F(\vec{x})] = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$F(\vec{x}) = \sum_{i=1}^{n} a_i F(\vec{v}_i) \quad \rightarrow \quad [F(\vec{x})] = \sum_{i=1}^{n} a_i [F(\vec{v}_i)]$$

Linear maps can be represented by matrices

- Coordinate vectors link generalized vector spaces with ordered lists:

$$F(\vec{x}) = \sum_{i=1}^{n} a_i F(\vec{v}_i) \quad \rightarrow \quad [F(\vec{x})] = \sum_{i=1}^{n} a_i [F(\vec{v}_i)]$$

- The column view of a linear system gives:

$$\sum_{i=1}^{n} a_i [F(\vec{v}_i)] = \begin{bmatrix} [F(\vec{v}_1)] & [F(\vec{v}_2)] & \cdots & [F(\vec{v}_n)] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = [F][\vec{x}]$$

- The matrix representation of a linear map is a matrix whose columns are the coordinate vectors of the mapped basis!
- When generalized vector spaces are written in terms of coordinate vectors, linear mappings are just matrix multiplication; another view of $A\vec{x} = \vec{b}$!

$$[F(\vec{x})] = [F][\vec{x}] \leftrightarrow A\vec{x} = \vec{b}$$

The differential matrix for P^3

- **A simple basis of P^3:**

$$E = \{1, \quad x, \quad x^2, \quad x^3\}, f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \rightarrow [f(x)]_E = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

- **The derivatives of the basis functions:**

$$e_1(x) = 1, e'_1(x) = 0 \rightarrow [e'_1(x)]_E = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, e_2(x) = x, e'_2(x) = 1 \rightarrow [e'_2(x)]_E = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$e_3(x) = x^2, e'_3(x) = 2x \rightarrow [e'_3(x)]_E = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix}, e_4(x) = x^3, e'_4(x) = 3x^2 \rightarrow [e'_4(x)]_E = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

The differential matrix for P^3

- **The differential matrix:**

$$D = \left[\frac{d}{dx}\right] = [[e'_1(x)]_E \quad [e'_2(x)]_E \quad [e'_3(x)]_E \quad [e'_4(x)]_E] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- **We can find the derivative simply by performing matrix multiplication:**

$$D[f(x)]_E = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ 2a_2 \\ 3a_3 \\ 0 \end{bmatrix} = [f'(x)]_E \;\rightarrow f'(x) = a_1 + 2a_2x + 3a_3x^2$$

$$\frac{d}{dx}\left(a_0 + a_1x + a_2x^2 + a_3x^3\right) = a_1 + 2a_2x + 3a_3x^2$$

Unit 13.4

Changing the Basis

Coordinate vectors depend on the choice of basis.

- **Consider two different bases for the same vector space:**

$$S = \{\vec{s}_1, \quad \vec{s}_2, \quad \cdots \quad \vec{s}_n\}, \qquad T = \{\vec{t}_1, \quad \vec{t}_2, \quad \cdots \quad \vec{t}_n\}$$

$$V = span(S) = span(T)$$

$$\forall \vec{x} \in V, \vec{x} = \sum_{i=1}^{n} a_i \vec{s}_i = \sum_{i=1}^{n} b_i \vec{t}_i \,, \rightarrow [\vec{x}]_S = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, [\vec{x}]_T = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

- **How do we change between bases?**

Subspaces of $\mathbb{R}^n$.

- If V is an m-dimensional subspace of $\mathbb{R}^n$, then $\vec{s}_i$ and $\vec{t}_i$ are "traditional" vectors.

$$\forall \vec{x} \in V \subseteq \mathbb{R}^n, \vec{x} = \sum_{i=1}^{m} a_i \vec{s}_i = \sum_{i=1}^{m} b_i \vec{t}_i, \rightarrow [\vec{x}]_S = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}, [\vec{x}]_T = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\sum_{i=1}^{m} a_i \vec{s}_i = [\vec{s}_1 \quad \vec{s}_2 \quad \cdots \quad \vec{s}_m] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} = P[\vec{x}]_S, \qquad \sum_{i=1}^{m} b_i \vec{t}_i = [\vec{t}_1 \quad \vec{t}_2 \quad \cdots \quad \vec{t}_m] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = Q[\vec{x}]_T$$

$$\vec{x} = P[\vec{x}]_S = Q[\vec{x}]_T$$

Subspaces of $\mathbb{R}^n$.

$$\vec{x} = P[\vec{x}]_S = Q[\vec{x}]_T, \qquad P = [\vec{s}_1 \quad \vec{s}_2 \quad \cdots \quad \vec{s}_m], Q = \left[\vec{t}_1 \quad \vec{t}_2 \quad \cdots \quad \vec{t}_m\right]$$

- **Both P and Q (n-by-m) have linearly independent columns (full column rank).**
- **Both P and Q (n-by-m) have a left inverse:** $P^+P = Q^+Q = I_m$

$$P[\vec{x}]_S = Q[\vec{x}]_T \to [\vec{x}]_S = P^+Q[\vec{x}]_T, [\vec{x}]_T = Q^+P[\vec{x}]_S$$

- **There is always a nonsingular (invertible) matrix that can convert between bases (because P and Q have the same column space):**
$$B = Q^+P, B^{-1} = P^+Q \to [\vec{x}]_T = B[\vec{x}]_S, [\vec{x}]_S = B^{-1}[\vec{x}]_T$$

Change of basis matrix.

$$[\vec{x}]_T = B[\vec{x}]_S, \qquad [\vec{x}]_S = B^{-1}[\vec{x}]_T$$

- **The columns of B are just the coordinates of the vectors $S = \{\vec{s}_1, \quad \vec{s}_2, \quad \cdots \quad \vec{s}_n\}$ written relative to $T = \{\vec{t}_1, \quad \vec{t}_2, \quad \cdots \quad \vec{t}_n\}$.**
  - **The coefficients needed to write: $\vec{s}_i = a_{1i}\vec{t}_1 + a_{2i}\vec{t}_2 + \cdots + a_{ni}\vec{t}_n$**

$$B = [[\vec{s}_1]_T \quad [\vec{s}_2]_T \quad \cdots \quad [\vec{s}_n]_T], \qquad B^{-1} = \left[[\vec{t}_1]_S \quad [\vec{t}_2]_S \quad \cdots \quad [\vec{t}_n]_S\right],$$

- **Why? Because $[\vec{s}_i]_T = B[\vec{s}_i]_S$ and $[\vec{s}_i]_S$ is a vector with a single 1 (and all other elements 0), that "picks out" column i from B!**
- **Because we are using only coordinate vectors here, the same approach works perfectly well with generalized vectors as well!**

Changing the basis of a transformation matrix.

$$F: V \to U, \qquad [F(\vec{x})] = [F][\vec{x}] \leftrightarrow A\vec{v} = \vec{u}$$

- **If we want to change the basis of for $\vec{v} = [\vec{x}]$, we multiply by the change of basis matrix for V:**

$$\vec{v} = B_V \vec{v}'$$

- **If we want to change the basis of for $\vec{u} = [F(\vec{x})]$, we multiply by the change of basis matrix for U:**

$$\vec{u} = B_U \vec{u}'$$

$$A\vec{v} = \vec{u} \to AB_V\vec{v}' = B_U\vec{u}' \to B_U^{-1}AB_V\vec{v}' = \vec{u}'$$

$$A'\vec{v}' = \vec{u}', A' = B_U^{-1}AB_V$$

Similar matrices.

- When we have a transformation with the same input and output spaces, $F: V \to V$, we may wish to use the same basis on both sides.

$$A'\vec{v}' = \vec{u}', A' = B_U^{-1} A B_V \to A' = B^{-1} A B$$

- When two square matrices share this relationship, we say that they are *similar*; they represent the same linear transformation, relative to different bases.
- Compare with the eigenvalue decomposition:

$$A = X \Lambda X^{-1} \leftrightarrow \Lambda = X^{-1} A X$$

- The eigenvalue decomposition finds a basis for V (the eigenvector basis) in which the linear transformation is diagonal!
    - Not always possible using a single basis (not all matrices are diagonalizable).

Transformation matrices and the SVD.

- Moving back to the general case, $F: V \to U$:

$$A'\vec{v}' = \vec{u}', A' = B_U^{-1} A B_V \;\to A = B_U A' B_V^{-1}$$

- Compare with the singular value decomposition:

$$A = U\Sigma V^T, V^T = V^{-1}, U^T = U^{-1}$$

- The singular decomposition finds an orthonormal basis for V and a (separate) orthonormal basis for U (the singular vector bases) in which the linear transformation is diagonal.
- This can be done for any linear mapping!

Orthogonality and generalized vector spaces.

- The SVD gives orthonormal bases for both the input and output spaces (domain and codomain), but what does this mean if we are dealing with generalized vector spaces?
- When we begin with a matrix representation of a linear map, everything is already based on coordinate vectors.
- One type of orthogonality is the orthogonality of coordinate vectors:
  - Coordinate vectors are orthogonal to one another if and only if the dot products of the coordinate vectors are equal to one.
- We can apply Gram-Schmidt orthonormalization to coordinate vectors to find an orthonormal set.